

EIVIS ANTWI SARFO

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Portfolio: <https://elvis-sarfo.web.app/>

Summary

Highly motivated and results-driven *Full-Stack Software Engineer* and a *Machine Learning Enthusiast* with over 8+ years of experience and a proven track record of designing, developing, and deploying high-performance web and mobile applications. Expertise in .NET, PHP, JavaScript, Angular and React, coupled with a strong understanding of database management and cloud technologies. Passionate about leveraging technology to create innovative solutions that solve real-world problems. Seeking a challenging Software Engineering role where I can contribute to the development of cutting-edge software and contribute to a dynamic team.

Skills

Technical skills

- **Libraries & Frameworks:** .NET, WordPress, Node.js, React, Angular
- **Machine Learning:** TensorFlow.
- **Database Management:** SQL, MongoDB, Firebase
- **Cloud & DevOps:** Google Cloud Services, Docker, Kubernetes

Programming Languages: C#, PHP, Typescript and Python

Soft Skills: Leadership, Communication, Teaching, Mentoring and Attention to Detail

Work Experience

Senior .NET Developer,

March 2025 – 2026

Layercake (JAX Tyres & Auto) | Australia (Remote)

- Maintained and enhanced internal JAX Tyres & Auto systems supporting franchise operations, reporting, and business workflows.
- Led modernisation of legacy .NET Framework applications by migrating them to .NET Core / .NET 6, improving code quality and long-term maintainability.
- Refactored backend services, APIs, and data access layers following modern architectural best practices.
- Planned and executed migration of on-premises SQL Server databases to Microsoft Azure, improving availability, scalability, and backup strategies.
- Provided production support, troubleshooting, and system enhancements for critical enterprise platforms.

Senior IT Technician,

March 2025 – March 2026

AAMUSTED | Ghana

- Administered and maintained the university's official website (WordPress), ensuring high availability, security updates, and content integrity for thousands of daily users across academic and administrative departments.
- Supported and maintained the School Clinic Management System, ensuring reliable access to patient records, appointment data, and operational reports for healthcare staff.
- Provided technical support and system maintenance for the Student Portal, resolving performance, access, and data issues to ensure uninterrupted academic services for students and faculty.
- Collaborated with academic and administrative stakeholders to implement updates, fix defects, and improve the usability of core digital systems.

Full Stack Software Developer,

November 2022 – 2025

Kline and Company | United States of America (Remote)

- Refactored the full stack (Dotnet Core backend, SQL queries, and Typescript functions) of an internal spreadsheet application, reducing data saving time by 78% (from 57 seconds to 12.4 seconds), significantly improving data science team productivity.
- Played a key role in developing a new forecasting tool for clients (including L'Oréal Group), increasing forecasting accuracy by 50% and leading to 70% improvement in data-driven business decision making.
- Integrated data visualisation charts into Angular applications, enhancing user experience and data comprehension for the company's Intelligent Centre, resulting in an increase in user engagement.
- Improved Angular application performance by 20% by implementing efficient change detection strategies, creating a faster and more responsive user experience.

Co-Founder and Tech Lead

September 2020 – Present

FARMASYST Company Limited | Ghana

- Led the development of an offline-first, secure agricultural management system, securing a grant from MasterCard and Kosmos Innovation Centre and contributing to the UN Sustainable Development Goals.
- Developed and integrated a novel deep learning model (95% accuracy) for offline plant disease detection within the mobile app, empowering farmers and supervisors in remote areas.
- Implemented an offline caching strategy in the Flutter mobile app, enabling reporting and monitoring in areas with limited internet connectivity, increasing data collection by 60%.
- Designed and developed a web dashboard providing stakeholders with data analysis and reporting capabilities, resulting in 70% data-driven decision-making.
- Configured and integrated Google Cloud services, optimising real-time communication and large data synchronisation between subsystems, reducing latency by 30%.

Google Developer Student Clubs (GDSC) Solution Challenge Winner & Project Lead Mentee

September 2020 – June 2021

Google Developers | United States of America (Remote)

- Led the team after winning the GDSC Solution Challenge 2020 with the Emergency Response Assistant, a software project enabling users to report health emergencies to emergency units in Ghana.
- Collaborated with Google engineers and expects to re-engineer the software architecture, addressing security vulnerabilities and improving scalability to accommodate a growing user base and new features, increasing system capacity.
- Streamlined development processes and improved code maintainability by adopting a modern tech stack (Firebase, Node.js, React, TypeScript, and Clean Architecture), reducing code defects by 60%.

Education

Master of Philosophy in Information Technology

2022-2026

Akenten Appiah-Menka University of Skills Training and Entrepreneurial Development, Kumasi, Ghana

- Focusing on machine learning and homomorphic encryption.
- Relevant coursework: Advanced Software Engineering Principles, Advanced Database Management, Advanced Data Structures and Algorithms, Information Security.

Bachelor of Science in Information Technology Education

2017-2021

University of Education, Winneba | Kumasi, Ghana (now AAMUSTED)

- Graduated with Distinction (GPA: 3.91), recognised as Best Graduating Student in the Faculty of Applied Sciences and Mathematics Education, and Second-Best Graduating Student overall.
- Relevant coursework: Data Structures and Algorithms, Discrete Mathematics, Numerical Methods and Computations, Software Engineering, Distributed Systems.

Research Experience

- **MPhil Thesis:** “Plant disease detection model for mobile devices using a novel convolutional neural network and quantisation aware training”
- **Published:** “Enhancing Efficiency and Speed in CKKS Homomorphic Encryption Through Power-Of-Two Vector Padding”: Proposed a power-of-two vector padding technique to improve the computational efficiency and speed of CKKS homomorphic encryption. (<https://link.springer.com/article/10.1186/s43067-025-00276-z>)
- **Automating Network Programmability and Backup on Cisco Devices Using Python and Netmiko Library: A Case Study of Komfo Anokye Teaching Hospital LAN.** (<https://jcrinn.com/index.php/jcrinn/article/view/488/356>)
- **Performance Comparison for NoSQL and SQL Databases:** Conducted a comparative analysis of NoSQL and SQL databases, identifying key performance differences in efficiency, scalability, and data handling.

Software Projects

Nextpress | Open Source Project

Jan 2026

Github Repo: <https://github.com/Elvis-Sarfo/nextpress>

- Designed and architected a modern, reusable CMS framework inspired by WordPress but built for low-bandwidth and enterprise environments.
- Developed with Next.js (App Router) and TypeScript, integrating Prisma, Auth.js, TipTap, and

Tailwind/ShadCN UI to deliver schema-driven content modelling, multilingual publishing, SEO guardrails, and role-based workflows—enabling editors to manage scalable, SEO-safe content efficiently across diverse regions.

Course Registration Portal | Department of Interdisciplinary Studies, AAMUSTED | Kumasi

2023

Watch Demo: <https://www.youtube.com/watch?v=HRjhRT-1tis&t=345s>

- Developed a Course Registration Portal catering to over 10000 continuing students in the Interdisciplinary Department at AAMUSTED, streamlining the process for registering in a variety of courses.
- Applied proficiency in PHP for backend development, MySQL for efficient data management, and front-end technologies such as JavaScript, HTML, and CSS to create a user-friendly interface.
- Successfully delivered the solution that not only met the technological requirements but also significantly improved the efficiency and experience for students engaging in course registration for 3 academic years

Emergency Response Assistance | Google Solution Challenge Project

2021

Watch Demo: <https://www.youtube.com/watch?v=tzRS94nivzo>

- Designed a two-way system to reduce the response time of emergencies by allowing victims to report emergency cases with live GPS location for easy routing and tracking of victims and emergency response teams.
- Used Node.js for backend development, Firebase for efficient data management, Google Maps API for geo-computations and front-end technologies such as JavaScript, HTML, and CSS to create a user-friendly interface.

Awards & Honors

- Best Teaching Assistant: NUGS Awards, AMMUSTED (2022)
- Best Graduating Student, Faculty of Applied Sciences and Mathematics Education, University of Education, Winneba, Kumasi Campus (June 2021)
- Winner, Google Developer Student Clubs Solution Challenge 2020 (July 2020).

Watch Demo: <https://www.youtube.com/watch?v=tzRS94nivzo>